SAheart: selecting an SVM classifier

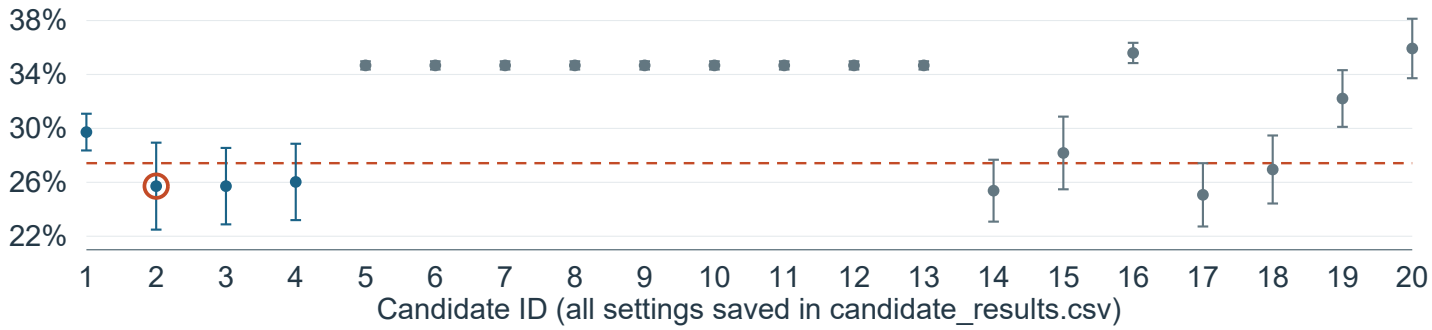
Data and design. 462 men; chd = 1 is positive (160 positive, 302 negative). Predictors: sbp, tobacco, ldl, adiposity, famhist, typea, obesity, alcohol, age; famhist: Absent = 1, Present = 2. No missing values or duplicate rows; nine numeric/coded predictors suit SVMs. No ID, group or time fields; assume independent men. Seed 432 gives training n = 323 (0/1: 211/112) and test n = 139 (91/48), rounding 30% within each class. Test labels stayed unused for tuning.

Prespecified analysis. Linear K(z,w) = sum_j z_j w_j tests a simple boundary; RBF K(z,w) = exp(-γ ||z-w||^2) allows smooth nonlinearity. Polynomial excluded: no specified interaction rationale. For both kernels C = {0.01, 0.1, 1, 10}; RBF γ = {0.001, 0.01, 0.1, 1}: 20 settings, no expansion. scikit-learn 1.9.1 SVC uses C as the summed hinge-loss penalty and γ as the squared-distance multiplier. Five shared stratified training folds; median imputation and numeric standardization (population SD) fitted within each analysis fold. Equal class weights; default zero-score boundary. Mean fold misclassification is primary for equal error costs and moderate imbalance.

Table 1. Kernel minima and selected model; errors and SEs in percent.

Kernel / candidate	C	γ	Mean CV	SE
Linear / 2 (selected)	0.1	-	25.72%	3.23%
RBF / 17	10	0.001	25.07%	2.35%

Figure 1. CV mean +/- SE: blue linear, gray RBF; ring selected; dashed cutoff.



Selection. SE = SD(fold errors)/sqrt(5); cutoff = 25.07% + 2.35% = 27.42%. Retain means at or below cutoff; prefer linear, then RBF, then smaller C/γ; remaining ties by ID. Selected linear, C = 0.1. Best linear minus minimum CV error is 0.64 percentage points; it does not exceed the minimum candidate's one SE (2.35 points). Correlated folds make SE a selection heuristic, not a confidence interval or significance test. These results do not clearly support nonlinear flexibility.

Final test. Frozen model refitted on all 323 training rows: 43/139 errors, error 30.94%, accuracy 69.06%; approximate binomial error-rate SE 3.92% assuming independent rows. Positive = chd 1: TN/FP/FN/TP = 72/19/24/24. Only 24/48 CHD cases detected; prediction is limited. Retrospective sampling and some post-treatment measurements limit prospective and population use; parameters are not causal effects.

Checks. Split/fold integrity, preprocessing leakage, one-SE arithmetic, parameters and confusion totals passed; all numbers regenerated from saved results. No analysis corrections required. Source: ESL SAheart notes; SVC kernel documentation.

Complete SKILL.md (unchanged appendix)

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name: fit-svm
description: Fit and compare SVM classifiers and produce a one-page report. Load only when explicitly invoked as fit-svm. Keep this file outside automatically discovered skill folders.

Fit SVM classifiers

1. Receive the data, response, predictors, categorical coding, and seed in the invocation; request missing essentials. Never invent results.
2. Check class counts, missingness, predictor types, identifiers, dependence, and sample size. Explain suitability for SVM classification; use grouped or temporal validation if independence is implausible, or explain why analysis cannot proceed.
3. For independent observations, reserve a stratified 30% test set using seed 432 unless specified. Set each class's test count to floor(0.30*n_class + 0.5); ensure both subsets retain every class. Record counts and indices; leave test outcomes unused until final evaluation.
4. On training data, fix five stratified CV folds shared by all candidates; reduce fold count if class sizes require it. Fit imputation, encoding, and numeric standardization within each analysis fold only; apply learned transformations to its validation fold.
5. Consider linear and radial-basis-function kernels to compare a simple boundary with a flexible nonlinear boundary. Include polynomial kernels only with a stated interaction or curvature rationale; explain inclusions and exclusions before tuning.
6. Tune C for every kernel, gamma for RBF, and degree/gamma/coef0 for polynomial if included. State the exact kernel formula and software parameterization. Prespecify a modest log-spaced C/gamma grid and a small degree grid; document any training-only grid expansion.
7. Prespecify mean CV misclassification rate as the primary criterion; use balanced error instead when class imbalance and the scientific objective warrant it. Fix class weights and decision rules using training data only; tune any weights or thresholds inside CV.
8. Save each candidate's fold errors, mean, and SE = SD(fold errors)/sqrt(number of folds). Note that correlated folds make this SE an approximate selection heuristic, not a confidence interval.
9. Use the one-SE rule: retain candidates with mean error <= minimum mean error + SE of that minimum. Prefer linear, then polynomial by increasing degree, then RBF; within a kernel prefer smaller C, then smaller gamma, then smaller coef0. Specify any remaining tie-break before tuning.
10. Compare the best linear candidate with the minimum-error candidate; state whether their CV gap exceeds the minimum candidate's one SE. Describe selection uncertainty without claiming statistical significance.
11. Freeze all choices, refit preprocessing and the selected SVM on all training rows, and evaluate once on the test set. Never revise the selected model or skill based on test performance.
12. Save one executable script, package versions, seed, split/fold indices, candidate results, selected parameters, and test predictions. Generate all reported numbers, figures, and tables from that run's saved results; do not copy numbers from earlier runs.

Report and verification

1.Produce at most one printed US Letter page with 11-point text, including figure/table labels; prioritize evidence for the final model.
2.Include data suitability, coding and split counts; kernel rationale; preprocessing, tuning grid and CV; selection rule and uncertainty; selected kernel/parameters; and test performance with limitations.
3.Use exactly one compact figure showing candidate CV errors with SE bars, the selected model, and the one-SE cutoff; fit the axis range to the estimates and bars so small differences remain visible.
4.Use exactly one compact table with each kernel's minimum-CV candidate and the selected model (deduplicate rows), tuning values, mean CV error, and SE; give test results only for the selected model.

5.Report test errors/total, error rate, accuracy, and approximate binomial error-rate SE when independent test observations justify it. Give binary confusion counts as TN/FP/FN/TP in prose, identifying the positive class; for multiclass, embed a labeled matrix panel in the single figure.
6.Interpret predictive performance and whether nonlinear flexibility is supported; do not treat SVM parameters as causal effects or ordinary regression coefficients.
7.Check leakage, split/fold integrity, selected parameters, one-SE arithmetic, confusion totals, and agreement of all numbers with saved results. Briefly state checks and any corrections actually made.
8.Render the report and verify one-page length, 11-point text, legibility, and no clipping; shorten content rather than shrink fonts.
9.Append the complete SKILL.md after the report, outside its one-page limit; attach the executable script and saved results separately without duplicating the invocation.